1. Project overview

Sprint 6 adds reproducible KMeans clustering, diagnostic outputs, and a read-only REST API for the Nifty100 analytics project.

Review note: retain actual command output and record any discrepancy before proceeding to the next gate.

2. Environment setup

Merge the supplied files into the existing N100 root. Activate the project virtual environment and install requirements-sprint6.txt. Keep the existing SQLite database.

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3. Database configuration

The API reads N100_DB when set, otherwise data/nifty100.db. Set N100_DB to an absolute path when running from a different directory. A missing database returns HTTP 503.

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4. Clustering workflow

Run python -m src.analytics.clustering. Missing feature values are imputed using broad-sector medians, then global medians, and finally zero if no numeric values exist.

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5. Clustering outputs

cluster_labels.csv contains company_id, cluster_id, cluster_name and distance_from_centroid. The elbow plot and inertia CSV show k values 2 through 10. Cluster archetype names are heuristic and need human review.

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6. Correlation and outliers

The pipeline produces a Pearson correlation heatmap for available core KPIs and flags sector-level absolute Z-scores greater than three. Small groups and zero-variance metrics are skipped.

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7. API startup

Run python -m uvicorn src.api.main:app --reload --port 8000 from the project root. Visit /docs for interactive API documentation and /openapi.json for the generated OpenAPI schema.

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8. Company data endpoints

GET /api/v1/companies lists companies with optional sector, market_cap_category and search filters. Company profile, P&L;, balance-sheet, cash-flow and ratio history routes use ticker symbols and return 404 for unknown tickers.

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9. Screener, sectors and peers

The screener supports ROE, debt/equity, FCF, sector, revenue CAGR, PAT CAGR and P/E filters. Check actual sector names and peer group names in the database before assuming a fixed count.

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10. Valuation and documents

Market-cap history returns valuation multiples by year. Company document endpoints include an is_url_valid flag based on URL scheme, not an external HTTP reachability check.

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11. Portfolio statistics

Portfolio stats returns P10, P25, P50, P75, P90, mean and standard deviation for available core metrics in the latest year.

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12. Testing

Run python -m pytest tests -v. The included tests are a smoke-test subset and do not claim the 60+ test exit criterion is met until the full existing project test suite is run.

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13. Performance

Measure ten concurrent screener calls and profile screen load times on the target machine. Record timings, hardware, database path, and test conditions in output/perf_notes.md.

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14. Troubleshooting

For import errors, run commands from the project root. For missing packages, install requirements-sprint6.txt in the active environment. For a missing database, inspect N100_DB and the current working directory.

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15. Security and deployment

CORS wildcard is included only for internal development per the sprint brief. Before production use, restrict allowed origins, deploy behind TLS, and add authentication and rate limits.

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16. Data quality

Run PRAGMA foreign_key_check and inspect row counts before sign-off. Treat null KPI values as unavailable rather than as financial zero unless the metric definition explicitly states otherwise.

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17. Acceptance gates

Use docs/acceptance_checklist.md and attach evidence for each gate. Gates involving manual calculation, visual PDF inspection, timings, and the screener comparison require actual local verification.

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18. API error handling

Expected responses include 404 for unknown tickers, sectors and peer groups; 400 for invalid screener values; and 503 when the database is unavailable.

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19. Daily review checklist

Check cluster counts, inspect the elbow curve, review the cluster profiles, smoke-test API routes, run the project test suite, and preserve logs with date/time and commit hash.

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20. Final sign-off

Archive the requested deliverables, confirm the checklist with evidence, and obtain the team lead signature and date. Do not mark any acceptance gate as passed based only on generated code.

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